

SYDNEY TECHNICAL HIGH SCHOOL



YEAR 12

MATHEMATICS EXTENSION 1

ASSESSMENT TASK TWO

2003

Time allowed: 70 minutes

Instructions

- * Attempt all questions.
- * Answers to be written on the paper provided.
- * Start each question on a new page.
- * Marks may not be awarded for careless or badly arranged working.

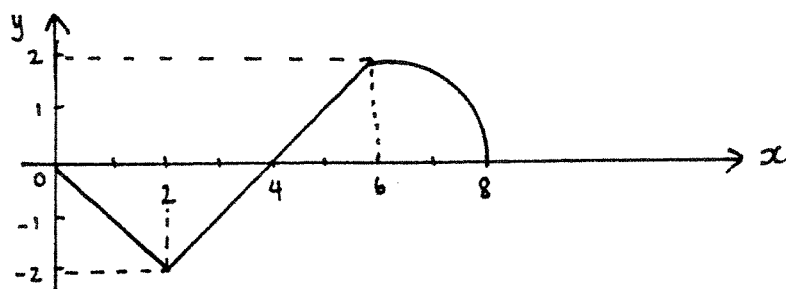
Q1	Q2	Q3	Q4	Q5	Q6	TOTAL
/9	/8	/9	/8	/9	/8	

Question 1**(9 Marks)**

a. Find i. $\int 4(1-2x)^3 dx$ ii. $\int (x^2-1)^2 dx$ (3)

b. Evaluate $\int_{-1}^4 \frac{x^2+x}{2x} dx$ (3)

c. The graph below shows the curve $y = f(x)$ between $x = 0$ and $x = 8$



Find the value of $\int_0^8 f(x) dx$ (1)

d. Evaluate $\int_{-4}^4 \sqrt{16-x^2} dx$ (2)

Question 2**(8 Marks)**

a. Evaluate $\int_{-2}^2 \frac{x^3}{1+x^2} dx$, giving a reason to support your answer (2)

b. Find the area bounded by the curve $y = x - x^2$, the x axis, and the lines $x = 1$ and $x = -1$ (3)

c. Find the value of $\int_0^2 \frac{dx}{\sqrt{1+2x}}$ (3)

Question 3**(9 Marks)**

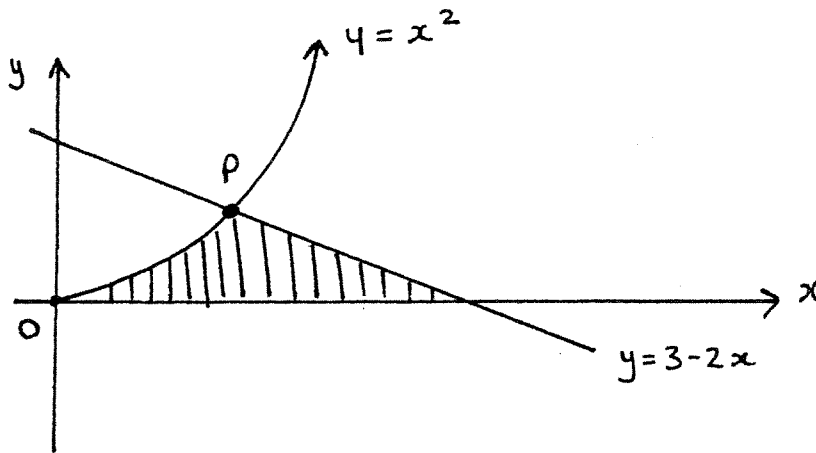
a. By use of the substitution $t = x + 1$

(4)

i.. Show $\int \frac{x}{\sqrt{x+1}} dx = \int (\sqrt{t} - \frac{1}{\sqrt{t}}) dt$

ii. Find the exact value of $\int_3^8 \frac{x}{\sqrt{x+1}} dx$

b.

(5)

The diagram shows the parabola $y = x^2$ and the line $y = 3 - 2x$ intersecting at the point P, in the first quadrant.

i. Show that P is the point (1, 1)

ii The shaded region is rotated about the x -axis. Find the volume of the solid formed.

Question 4**(8 Marks)**

a. i.. Estimate the area between the function $y = \sqrt{x}$, the x -axis and the ordinates $x=0$ and $x=1$, correct to 3 decimal places by using Simpson's rule with 3 function values.

(4)

ii Find the percentage error in (i) correct to two decimal places

b. The graphs $y = 16 - x^2$ and $y = 6x$ intersect at P and Q.

(4)

i. Find the x -values of P and Q

ii. Hence, find the area between the two curves

Questions 5.

(9 Marks)

a. Find $\int \frac{2x}{\sqrt{x^2 - 4}} dx$ using the substitution

(3)

$$u = x^2 - 4.$$

b. Consider the function $y = x + 2 + \frac{4}{x-1}$

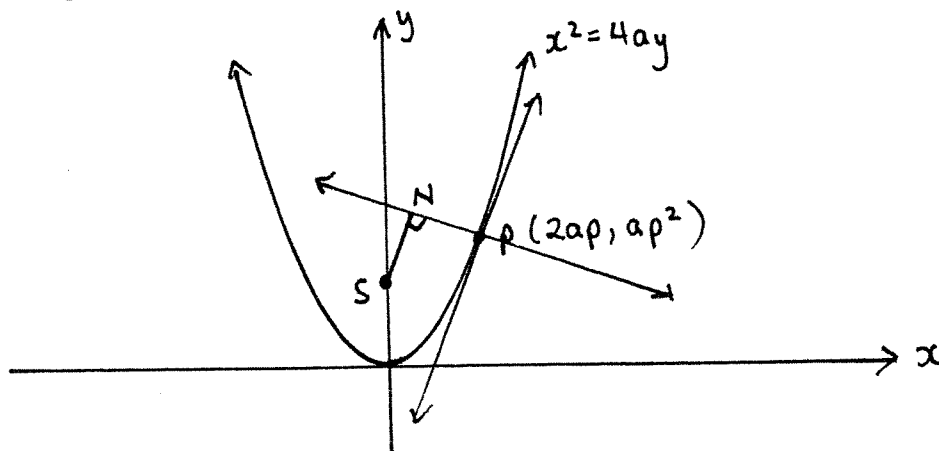
(6)

- For what values of x is the function underfined ?
- What is the equation of the oblique asymptote ?
- Find the co-ordinates of any stationary points and determine their nature
- Sketch the curve $y = x + 2 + \frac{4}{x-1}$.

Question 6

(8 Marks)

$P(2ap, ap^2)$ is a point on the parabola at $x^2 = 4ay$. SN is perpendicular to the normal at P, where S is the focus of the parabola and N the foot of the perpendicular from SN to the normal



- a. Show that the equation of the normal at P is

$$x + py = 2ap + ap^3$$

- b. Find the equation of SN

- c. Show that the co-ordinates of the point N are $(ap, ap^2 + a)$

- d. Find the locus of N as P moves on the parabola

Question 1

$$\begin{aligned} \therefore \text{ i. } \int 4(1-2x)^3 dx &= 4 \int 2(1-2x)^3 dx \\ &= -2(1-2x)^4 \\ &= -\frac{1}{2}(1-2x)^4 + C \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{ii. } \int (x^2-1)^2 dx &= \int x^4 - 2x^2 + 1 dx \quad \checkmark \\ &= \frac{x^5}{5} - \frac{2}{3}x^3 + x + C \quad \checkmark \end{aligned}$$

* No mark unless +C attached

$$\begin{aligned} \text{3. } \int_{-1}^4 \frac{x^2+x}{2x} dx &= \frac{1}{2} \int_{-1}^4 \frac{x^2}{x} + \frac{x}{x} dx \quad \checkmark \\ &= \frac{1}{2} \left[\frac{x^2}{2} + x \right]_{-1}^4 \quad \checkmark \\ &= \frac{1}{2} \left[\frac{16}{2} + 4 - \left(\frac{1}{2} - 1 \right) \right] \\ &= 6\frac{1}{4} \quad \checkmark \end{aligned}$$

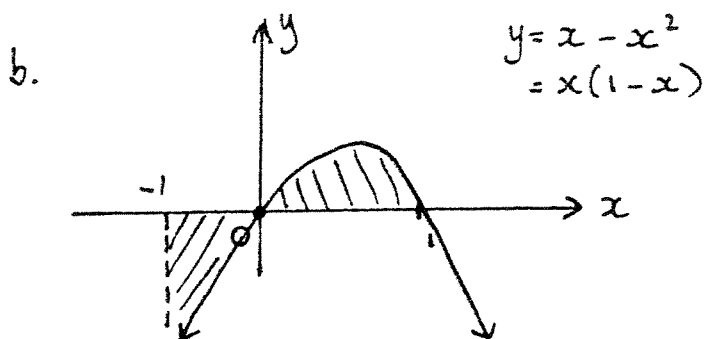
$$\begin{aligned} \therefore \int_0^8 f(x) dx &= -\frac{1}{2} \times 2 \times 4 + \frac{1}{2} \times 2 \times 2 + \frac{1}{4} \times \pi \times 2^2 \\ &= -4 + 2 + \pi \\ &= \pi - 2 \quad \text{(incorrect if units such as } u^2 \text{ written)} \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{i. } \int_{-4}^4 \sqrt{16-x^2} dx &= \frac{1}{2} \text{ circle (radius 4)} \\ &= \frac{1}{2} \times \pi \times 4^2 \quad \checkmark \\ &= 8\pi \quad \checkmark \end{aligned}$$

Question 2

$$\text{a. } \int_{-2}^2 \frac{x^3}{1+x^2} dx = 0 \quad \checkmark$$

odd fnc between symmetrical limits. ✓



$$\begin{aligned} A_1 &= \int_0^1 x - x^2 dx \\ &= \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 \\ &= \frac{1}{2} - \frac{1}{3} - 0 \\ &= \frac{1}{6} \quad \checkmark \end{aligned}$$

$$\begin{aligned} A_2 &= \left| \int_{-1}^0 x - x^2 dx \right| \quad \text{or } \int_0^{-1} \\ &= \left| \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_{-1}^0 \right| \\ &= \left| 0 - \left(\frac{1}{2} - \frac{1}{3} \right) \right| \\ &= \left| -\frac{1}{6} \right| \\ &= \frac{1}{6} \quad \checkmark \end{aligned}$$

$$\therefore \text{ Area} = \frac{1}{6} + \frac{1}{6} = \frac{1}{3} \quad \checkmark$$

$$\begin{aligned} \text{c. } \int_0^2 (1+2x)^{-1/2} dx &= \frac{(1+2x)^{1/2}}{\frac{1}{2} \times 2} \quad \checkmark \\ &= \left[\sqrt{1+2x} \right]_0^2 \quad \checkmark \\ &= \sqrt{5} - \sqrt{1} \\ &= \sqrt{5} - 1 \quad \checkmark \end{aligned}$$

Question 3

$$\int \frac{x}{\sqrt{x+1}} dx \quad t = x+1$$

$$\frac{dt}{dx} = 1$$

$$dx = dt$$

$$\int \frac{t-1}{\sqrt{t}} dt \quad \checkmark$$

$$\int \frac{t}{\sqrt{t}} - \frac{1}{\sqrt{t}} dt \quad \checkmark$$

$$\int \sqrt{t} - \frac{1}{\sqrt{t}} dt$$

$$\int_3^8 \frac{x}{\sqrt{x+1}} dx \quad x=8 \quad t=9$$

$$x=3 \quad t=4$$

$$\int_4^9 \sqrt{t} - \frac{1}{\sqrt{t}} dt \quad \checkmark$$

$$= \left[\frac{2}{3} t^{3/2} - 2t^{1/2} \right]_4^9$$

$$\frac{2}{3} \sqrt{9}^3 - 2\sqrt{9} - \left(\frac{2}{3} \sqrt{4}^3 - 2\sqrt{4} \right) \quad \checkmark$$

$$18 - 6 - (1\frac{1}{3})$$

$$= 10\frac{2}{3}$$

$$1. \quad x^2 = 3 - 2x$$

$$x^2 + 2x - 3 = 0$$

$$(x+3)(x-1) = 0$$

$$x = 1 \quad y = 1$$

$$\therefore P(1, 1) \quad \checkmark$$

$$(II) \quad V_x = \pi \int y^2 dx$$

$$V_1 = \pi \int_0^1 x^4 dx \quad \checkmark$$

$$= \pi \left[\frac{x^5}{5} \right]_0^1$$

$$= \pi \left[\frac{1}{5} - 0 \right]$$

$$= \pi/5$$

$$V_2 = \pi \int_1^{3/2} (3-2x)^2 dx$$

$$= \pi \left[\frac{(3-2x)^3}{-6} \right]_1^{3/2} \quad \checkmark$$

$$= -\frac{\pi}{6} \left[0 - (3-2)^3 \right]$$

$$= -\frac{\pi}{6} \times -1$$

$$= \pi/6 \quad \checkmark$$

$$\therefore \text{Volume} = \frac{11\pi}{30} \quad u^3 \quad \checkmark$$

Question 4.

x	0	$\frac{1}{2}$	1
$\sqrt{x}$	0	$\frac{1}{\sqrt{2}}$	1

$$1 \div \frac{h}{3} (f + L + 4m)$$

$$= \frac{1}{6} \left(0 + 1 + \frac{4}{\sqrt{2}} \right) \checkmark$$

$$= 0.63807 \dots$$

$$= 0.6381 \text{ (4 sig fig)} \checkmark$$

$$\int_0^1 \sqrt{x} \, dx$$

$$= \left[\frac{2}{3} x^{3/2} \right]_0^1$$

$$= \frac{2}{3} - 0$$

$$= \frac{2}{3} \checkmark$$

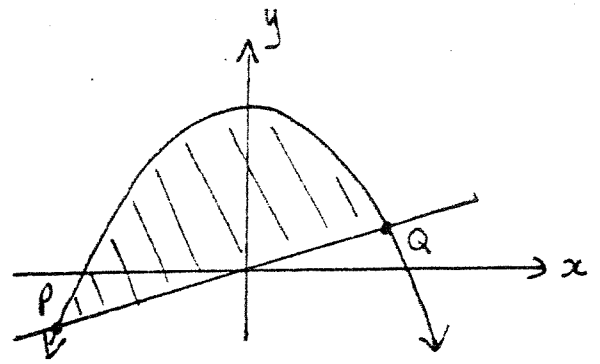
$$\% \text{ error} = \frac{\text{diff}}{\text{exact}} \times 100\%$$

$$= \frac{0.02859}{2/3} \times 100\%$$

$$= 4.289 \dots$$

$$= 4.29\% \checkmark$$

b.



$$1. \quad 6x = 16 - x^2$$

$$x^2 + 6x - 16 = 0$$

$$(x + 8)(x - 2) = 0$$

$$x = -8 \quad x = 2 \checkmark$$

$$11. \quad A = \int_{-8}^2 (16 - x^2) - 6x \, dx \checkmark$$

$$= \left[16x - \frac{x^3}{3} - 3x^2 \right]_{-8}^2 \checkmark$$

$$= 32 - \frac{8}{3} - 12 - \left(-128 + \frac{8^3}{3} - 3 \times 64 \right)$$

$$= 17\frac{1}{3} - (-149\frac{1}{3})$$

$$= 166\frac{2}{3} \checkmark$$

Question 5

$$\int \frac{2x}{\sqrt{x^2-4}} dx \quad u = x^2 - 4$$

$$\frac{du}{dx} = 2x$$

$$\int \frac{1}{\sqrt{x^2-4}} 2x dx \quad du = 2x dx$$

$$= \int \frac{1}{\sqrt{u}} du$$

$$= \int u^{-1/2} du$$

$$= 2u^{1/2}$$

$$= 2\sqrt{x^2-4} + C$$

o.

i. undefined at $x = 1$

at $x = 3 \quad y'' > 0 \therefore \text{Min}$
(3,7)

ii. $y = x + 2$

at $x = -1 \quad y'' < 0 \therefore \text{Max}$
(-1,-1)

∴ $dy/dx = 0$ stat pts

$$\frac{dy}{dx} = 1 - 4(x-1)^{-2}$$

$$0 = 1 - \frac{4}{(x-1)^2}$$

$$\frac{4}{(x-1)^2} = 1$$

$$4 = (x-1)^2$$

$$x-1 = 2 \quad x-1 = -2$$

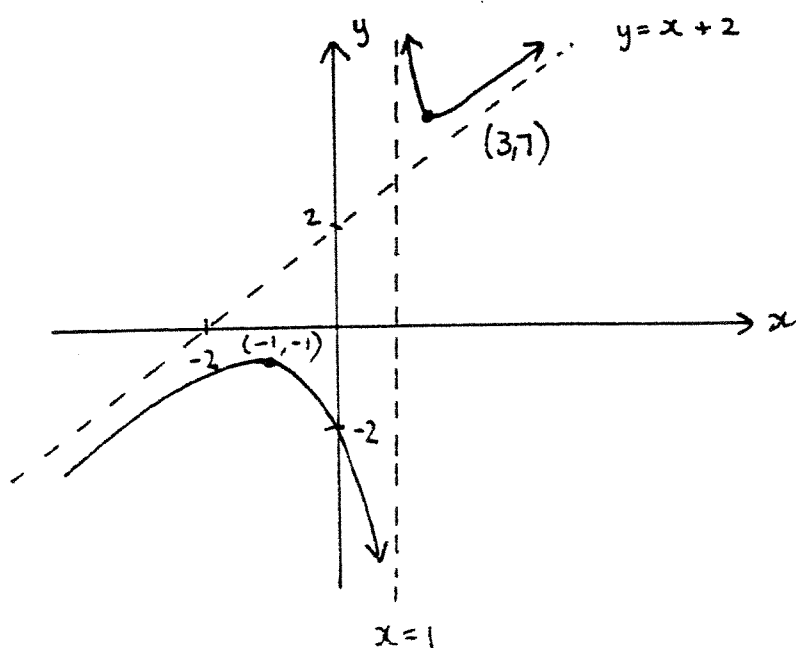
$$x = 3$$

$$x = -1$$

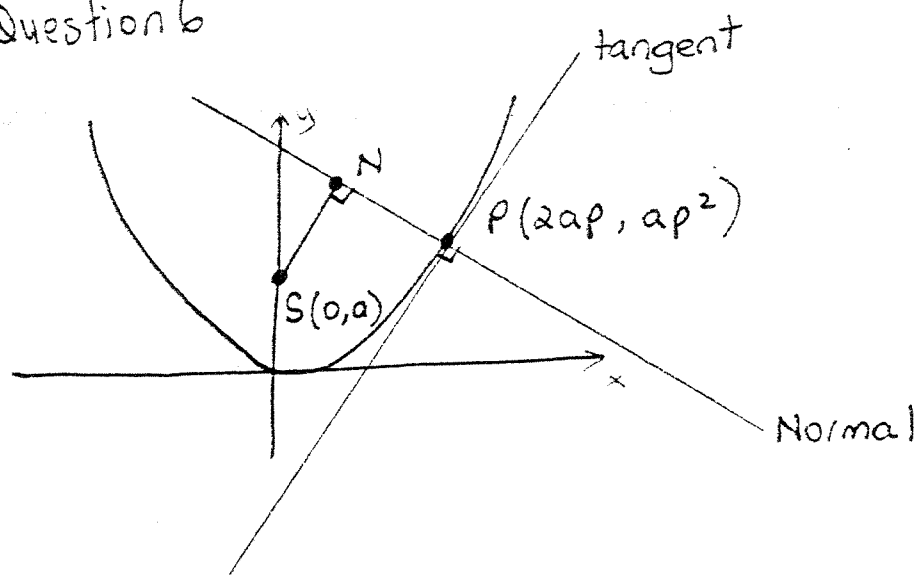
$$y = 7$$

$$y = -1$$

$$\frac{d^2y}{dx^2} = \frac{8}{(x-1)^3}$$



Question 6



$$x = ap$$

$$\begin{aligned} y &= p(ap) + a \\ &= \checkmark ap^2 + \checkmark a \end{aligned}$$

$$\text{ie } N(ap, ap^2 + a)$$

i. $x^2 = 4ay$

$$\frac{x^2}{4a} = y$$

$$y' = \frac{2x}{4a} \text{ at } x = 2ap$$

$$m_T = p$$

$$\therefore m_N = -\frac{1}{p} \quad \checkmark$$

$$\text{ie } y - ap^2 = -\frac{1}{p}(x - 2ap)$$

$$py - ap^3 = -x + 2ap$$

$$x + py = 2ap + ap^3 \quad \checkmark$$

ii. SN $m_{SN} = p \quad \checkmark$

$$\begin{aligned} \therefore y - a &= p(x - 0) \\ y &= px + a \quad \checkmark \end{aligned}$$

iii. $x + py = 2ap + ap^3$
 $y = px + a$

$$\begin{aligned} \therefore x + p(px + a) &= 2ap + ap^3 \\ x + p^2x + ap &= 2ap + ap^3 \\ x(1 + p^2) &= ap + ap^3 \\ x(1 + p^2) &= ap(1 + p^2) \end{aligned}$$

IV Locus

$$x = ap \quad \checkmark$$

$$\frac{x}{a} = p \quad \therefore y = ap^2 + a$$

$$\checkmark y = a\left(\frac{x}{a}\right)^2 + a$$

any pt
from
here on
correct

$$\downarrow y = a \cdot \frac{x^2}{a^2} + a$$

$$y = \frac{x^2}{a} + a$$

$$ay = x^2 + a^2$$

$$x^2 = ay - a^2$$

$$x^2 = a(y - a)$$